Scripts and raw results available at

Bayesian Stage-Fall-Discharge Laws and Effective River Depth Estimation from SWOT Altimetry and Slope

Ngoc Bao NGUYEN(1), Pierre-André GARAMBOIS(1), Benjamin RENARD(1), Kevin LARNIER(2), Ngo Nghi Truyen HUYNH(1)

Affiliation: INRAE RECOVER (1), HYDROMATTER (2)

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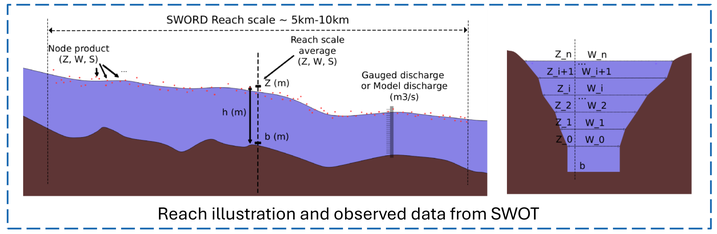
Context - Inverse Hydraulics from River Observations

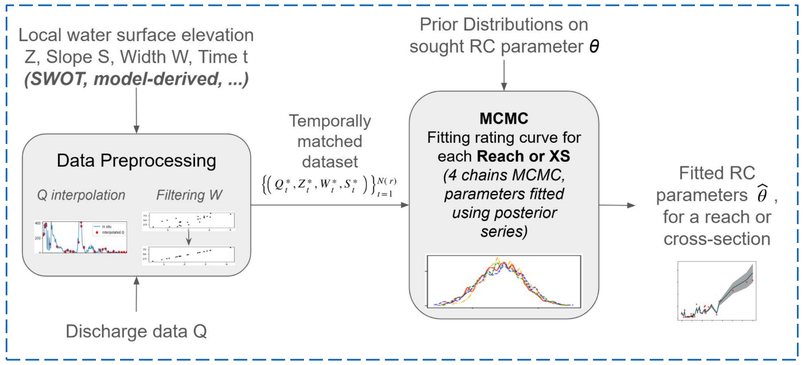
- **Stage-discharge rating curves (RCs) calibration relies on in situ gaugings:** costly, sparse, hard to sustain in remote basins (Le Coz et al., 2014; McMillan et al., 2017; Fenton, 2018)
- **SWOT observation provides hydraulic visibility of river surface globally:** spatially distributed water surface elevation (WSE), width, slope (Garambois et al., 2017; Rodriguez et al., 2020)
- **Discharge inversion remains ill-posed** (non-unique flux-from-geometry Q from Z, W, S alone), sensitivity to prior and measurement (Garambois et al., 2015; Larnier et al., 2020; Larnier et al., 2023)



Reach illustration and observed data from SWOT

$$\mathcal{F}^{lFM}: \quad Q(Z,S,W) = K_3\Big(W_0(Z_0-b_3)+\delta A(Z)\Big)^{5/3} W^{-2/3} S^{1/2}$$

Methodology – Bayesian Rating Curve estimation at reach scale or node scale



Discharge laws of increasing complexity

$$\mathcal{F}^{RC}: \quad Q(Z) = \alpha_1 (Z-b_1)^{\beta_1},$$
$$\mathcal{F}^{SFD}: \quad Q(Z,S) = \alpha_2 (Z-b_2)^{\beta_2} S^{\delta_2},$$
$$\mathcal{F}^{lFM}: \quad Q(Z,S,W) = K_3\Big(W_0(Z_0-b_3)+\delta A(Z)\Big)^{5/3} W^{-2/3} S^{1/2}.$$

Data and scale: method applicable at a section or reach resolution, from model or observations

- Reach-scale or node scale observations: WSE Z, slope S, width W
- Reference discharge from:
  - in situ gauges (gauged reaches)
  - GRADES-HydroDL (pseudo-ungauged reaches) (Yang et al. 2024)
  - hydraulic model outputs (noise-free benchmark case)
- Real data large-scale experiment: ~4,847 reaches total (including ~4,013 ungauged)

Inference approach

- Bayesian calibration with MCMC (*PyMC*), per reach
- Error model combines observation and structural uncertainty $\quad Q_t^* = \mathcal{F}^{\square}(y_t^*|\boldsymbol{\theta}) + \epsilon_t + \eta_t$
- Outputs:
  - posterior distributions of law parameters
  - inferred effective bathymetry/depth
  - discharge estimates and uncertainty bands

Structural error

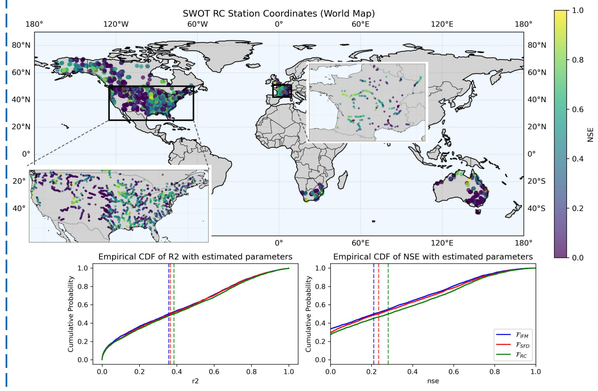
Observation error

Results

Global rating curves from reach scale swot and reference discharge from hydrological models or in situ

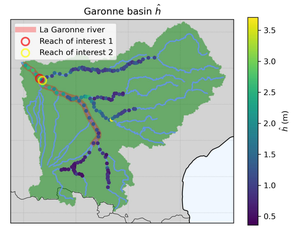
Using SWOT observations with hydrological model reference discharge, performance is encouraging:

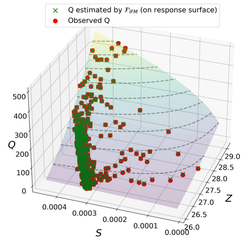
- About 25% of reaches achieve strong skill (NSE>0.7, R^2>0.7)
- Median performance is moderate (around NSE≈0.28, R^2≈0.38)
- Error sources might come from SWOT processing, hydrological regionalization, applicability range of discharge law, data preprocessing



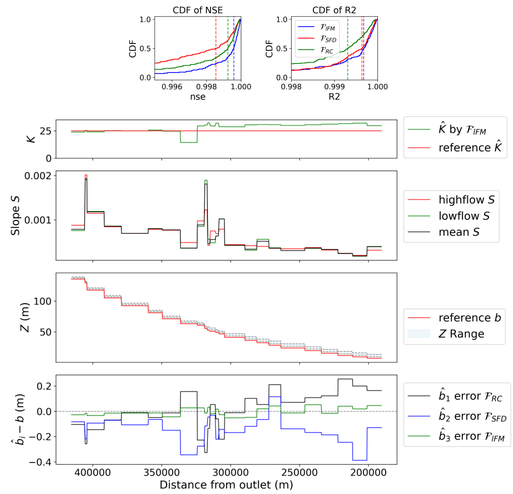
Rating curves derived from network-scale hydrological-hydraulic model outputs, spatial patterns of RC parameters

- **Controlled experiment (noise-free, physically consistent inputs: Saint Venant river network model with swot based cross sections):** near-perfect performance, confirming internal consistency and robustness of the Bayesian framework, capture hysteresis
- **Spatial consistency** of inferred parameters





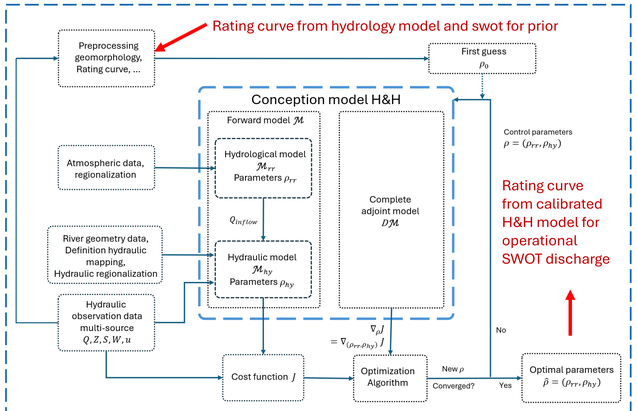
Discharge law IFM response surface on reach of interest 2, **stage-discharge hysteresis captured with slope**



Inferred mean river depth and error spatial patterns on the Garonne river main stem

Conclusions & Perspectives

- **Feasible and scalable Bayesian SWOT-based discharge law calibration**: reach to network scale; gauged + ungauged contexts
- **Unified probabilistic outputs**: discharge estimates, uncertainty bounds, effective bathymetry priors
- **Main limitation**: input data quality, especially noisy SWOT slope and width
- **Priority next steps**: improved SWOT denoising, stronger hydrology-hydraulic data assimilation coupling (basin-scale closure), spatially constrained/regionalized inference over river networks
- **Towards** RCs chainage (prior, model derived) with full SMASH-DassFlow H&H adjoint (flwg Pujol et al. 2022, Pujol et al. Submitted 2026, Ettalbi et al. In prep.), enriched HiVDI for network scale inversions, learning of hybrid hydrology model(s), of hydraulic correction NN for bathymetry-friction globally; surrogates, UQ



Complete integrated Hydrology-Hydraulic model conception

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